AJOU UNIVERSITY
DEPT. OF ELECTRICAL AND COMPUTER ENGINEERING

Homework #1

Subheading: *problems on discrete random variables*

Course: *Probability and Random Variables* – Professor: *Bang Chul Jung*
Due Date: *15:00 April 13th, 2026 (before the class start)*

Problem 1

The random variable V has PMF:

$$P_V(v) = \begin{cases} cv^2 & v = 1, 2, 3, 4, \\ 0 & \text{otherwise.} \end{cases}$$

(a) Find the value of the constant c.

(b) Find $P\left[V \in \{u^2 | u = 1, 2, 3, \cdots\}\right]$.

(c) Find the probability that V is even.

(d) Find $P[V > 2]$.

Problem 2

In each at-bat in a baseball game, mighty Casey swings at every pitch. The result is either a home run (with probability $q = 0.05$) or a strike. Of course, three strikes and Casey is out.

(a) What is the probability P$[H]$ that Casey hits a home run?

(b) For one at-bat, what is the PMF of N, the number of times Casey swings his bat?

Problem 3

Suppose when baseball player gets a hit, a single is twice as likely as a double, which is twice as likely as a triple, which is twice as likely as a home run. Also, the player's batting average, i.e., the probability the player gets a hit, is 0.300. Let B denote the number of bases touched safely during an at-bat. For example, $B = 0$ when the player makes an out, $B = 1$ on a single, and so on. What is the PMF of B?

Problem 4

A packet received by your smart phone is error-free with probability 0.95, independent of any other packet. In one hour, your smartphone receives $12,000$ packets. Let X equal the number of packets received with errors. What is E$[X]$?

Problem 5

It can take up to four days after you call for service to get your computer repaired. The computer company charges for repairs according to how long you have to wait. The number of days D until the service technician arrives and the service charge C, in dollars, are described by

d	1	2	3	4
$P_D(d)$	0.2	0.4	0.3	0.1

and

$$C = \begin{cases} 90 & \text{for 1-day service} \\ 70 & \text{for 2-day service} \\ 40 & \text{for 3-day service} \\ 40 & \text{for 4-day service} \end{cases}$$

(a) What is the expected waiting time $\mu_D = \mathrm{E}[D]$?

(b) What is the expected deviation $\mathrm{E}[D - \mu_D]$?

(c) Express C as a function of D.

(d) What is the expected value $\mathrm{E}[C]$?

(e) What is the standard deviation σ_D of the waiting time?

Problem 6

X is the binomial $(5, 1/2)$ random variable. Find $P_{X|B}(x)$, where the condition $B = \{X \geq \mu_X\}$. What are $\mathrm{E}[X|B]$ and $\mathrm{Var}[X|B]$?

Problem 7

X and Y are independent identical discrete uniform $(1, 10)$ random variables. Let B denote the event that $\max(X, Y) \leq 5$. Find the condition PMF $P_{X,Y|B}(x, y)$.

Problem 8

N and K have joint PMF

$$P_{N,K}(n,k) = \begin{cases} \frac{(1-p)^{n-1}p}{n} & \begin{array}{l} n = \; 1,2,... \\ k = \; 1,...,n, \end{array} \\ 0 & \text{otherwise.} \end{cases}$$

Let B denote the event that $N \geq 10$.

(a) Find the conditional PMFs $P_{N|B}(n)$ and $P_{N,K|B}(n,k)$. Which should you find first?

(b) Find the conditional expected value $\mathrm{E}[N|B]$, $\mathrm{E}[K|B]$, $\mathrm{E}[N+K|B]$, $\mathrm{Var}[N|B]$, $\mathrm{Var}[K|B]$.

Problem 9

Select integrated circuits, test them in sequence until you find the first failure, and then stop. Let N be the number of tests. All tests are independent, with probability of failure $p = 0.1$. Consider the condition $B = \{N \geq 20\}$.

(a) Find the PMF $P_N(n)$.

(b) Find $P_{N|B}(n)$, the conditional PMF of N given that there have been 20 consecutive tests without a failure.

(c) What is $\mathbb{E}[N|B]$, the expected number of tests given that there have been 20 consecutive tests without a failure.

Problem 10

X is the binomial (5,0.5) random variable.

(a) Find the standard deviation of X

(b) What is $P[\mu_X - \sigma_x \leq X \leq \mu_X + \sigma_x]$, the probability that X is within one standard deviation of the expected value?